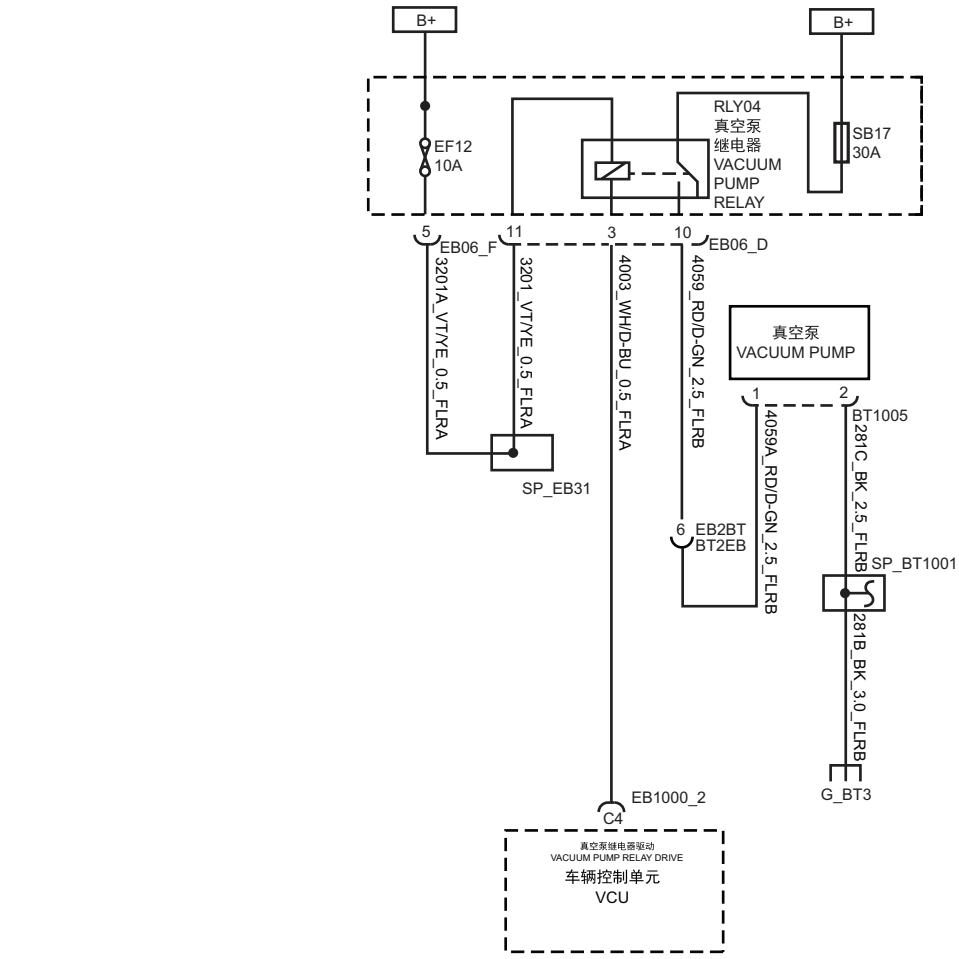
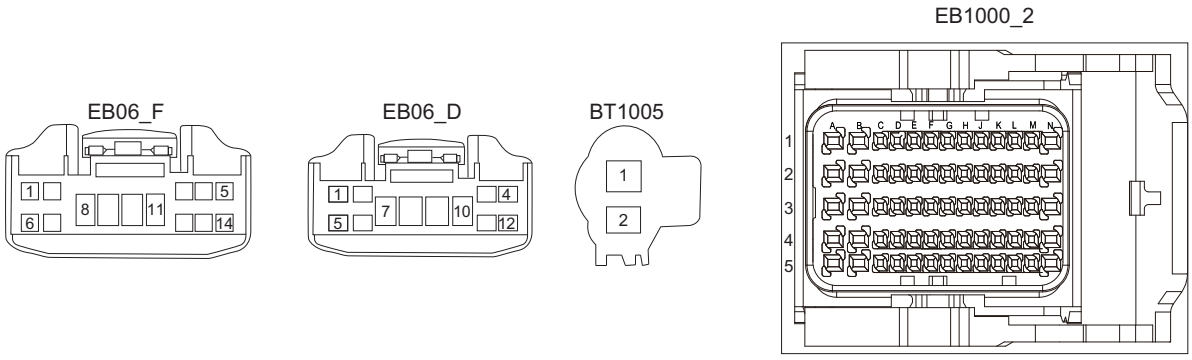
P258D、 P258C、 P258A

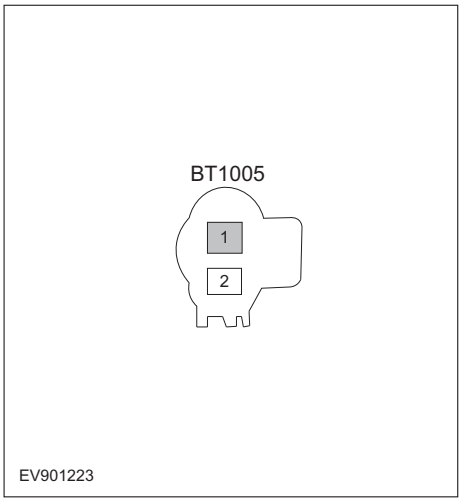
Fault diagnosis code
P258D：DEM_stVacPmpSBCfmd
P258C：DEM_stVacPmpSGCfmd
P258A：DEM_stVacPmpOCCfmd
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • VCU module fault • The brake vacuum pump fault.
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P258D
The underlying diagnosis
P258C
The underlying diagnosis
P258A
The underlying diagnosis
To set the effect of a fault code condition
Description of circuit diagram
The VCU module monitors the brake vacuum pump via low level reference circuit and signal circuit.

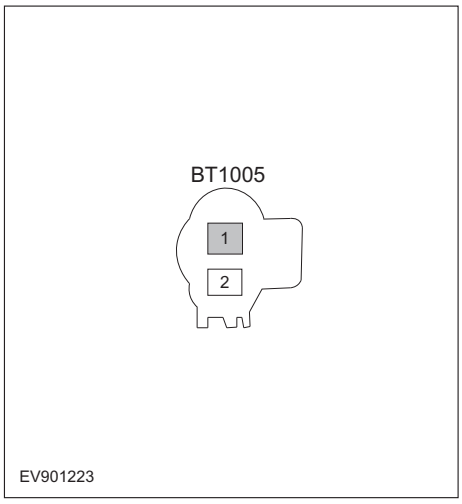
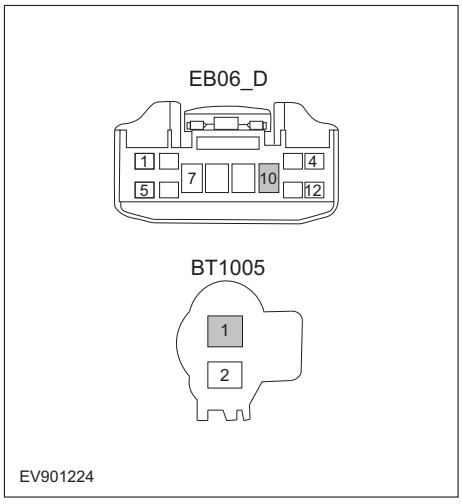
Circuit diagram

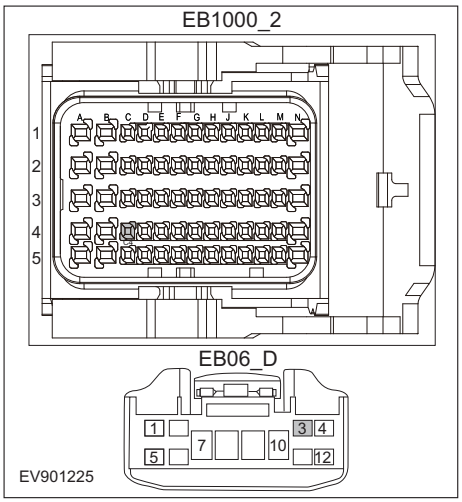


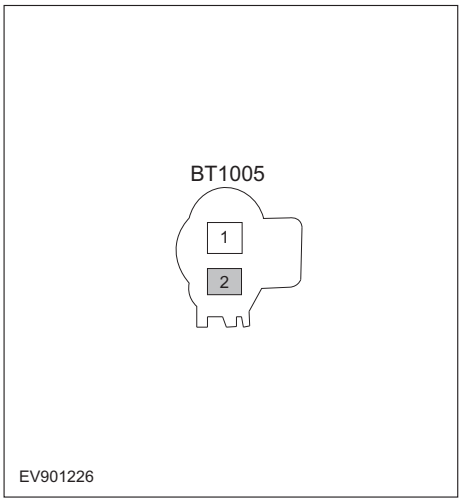


EV901204

Troubleshooting steps	
Step 1	General inspection.
	1 Check fuses SB17/EF12 for damage. 2 Check the relay RLY04 for damage. 3 Check battery capacity. 4 Check the harness plug of brake vacuum pump,VCU for damage, poor contact, aging and looseness. 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 2	Check fault codes.
	1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the VCU with diagnostic apparatus. 4 Is there any other fault code except for P258D/P258C/P258A?
	YES → Refer to:DTC Summary list(VCU).
NO ↓	
Step 3	Check the power voltage of brake vacuum pump battery.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug BT1005 of brake vacuum pump . 3 Turn the ignition switch to ON. 4 Measure the voltage between the plug terminal of brake vacuum pump and ground. Measuring circuit: voltage between terminal 1 on plug BT1005 and ground. **Standard value: 0V** 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 4	Check the supply power of brake vacuum pump when it is started or operating.
	1 Turn the ignition switch to ON. 2 Step down the brake pedal and make the brake vacuum pump relay in working state. 3 Measure the voltage between the plug terminal of brake vacuum pump and ground. Measuring circuit: voltage between terminal 1 on plug BT1005 and ground. **Standard value:10~14V** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 5	Check the circuit between brake vacuum pump relay and brake vacuum pump(open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB06_D of UEC. 3 Measure the resistance between the harness terminal of UEC and the brake vacuum pump. Measuring circuit: resistance between terminal 10 on plug EB06_D and terminal 1 on plug BT1005. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Check the circuit between brake vacuum pump relay and VCU (open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB1000_2 of VCU module. 3 Measure the resistance between the harness terminal of UEC and the VCU. Measuring circuit: resistance between terminal 3 on plug EB06_D and terminal C4 on plug EB1000_2. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 7	Check the grounding circuit of brake vacuum pump .
	1 Turn the ignition switch to OFF. 2 Measure the resistance between the harness plug of brake vacuum pump and ground. Measuring circuit: resistance between terminal 2 on plug BT1005 and ground. **Standard value: ≈ 0Ω** 3 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 8	Check the brake vacuum pump relay . (components)
	1 Turn the ignition switch to OFF. 2 Replace the relay of brake vacuum pump in good condition. 3 Connect the fault diagnosis instrument to the diagnosis interface. 4 Turn the ignition switch to ON position. 5 Clear fault codes. 6 Reread fault codes. 7 Test whether the faulted code is existed.
	NO → Replace the relay of brake vacuum pump.
YES ↓	
Step 9	Test whether VCU is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB1000_1, IP55_A of VCU. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the VCU plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Change the VCU Module
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.	
Next Step ↓	
Diagnosis end.	